UNIVERSITI TEKNOLOGI PETRONAS

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Project Report: USV Marine Plastic Collection Simulation

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USV Marine Plastic Collection: Discrete Agent-Based Modelling and Cellular Automata

1. Introduction

This NetLogo model demonstrates a small part of the proposed USV Marine Plastic Collection System, specifically Scenario 1: Normal Plastic Collection. The ABM component represents the autonomous behaviour and interaction of two Unmanned Surface Vehicles (USVs), while the cellular automata component represents the changing water surface.

2. Discrete Agent-Based Model

The discrete agent-based component models the system as individual autonomous agents that make decisions based on their own state and surrounding conditions. In this simulation, the two active agents are the Search-and-Collect USV (SC-USV) and the Transport USV (T-USV). The SC-USV changes between states such as searching, waiting and completed, while the T-USV changes between idle, retrieving and returning. These state changes occur at discrete simulation ticks, which makes the model suitable for representing step-by-step decisions and interactions between autonomous vehicles.

The importance of using Discrete ABM is that each USV can behave independently while still contributing to a shared task. The SC-USV patrols the environment, detects and collects plastic, while the T-USV responds when collection storage becomes full and transports the waste back to the base. This allows the model to demonstrate important agent-based concepts such as goal seeking, sensing, role specialisation, state transitions, communication and cooperation, instead of modelling the whole system as one single process.

3. Cellular Automata Model

The Cellular Automata component is used to represent the water surface as a grid of individual cells. In NetLogo, each patch acts as one CA cell and stores values such as the current wave height and wave energy. At every simulation tick, each cell updates its next state using its own current value together with the values of its eight neighbouring cells. Because all cells follow the same local update rule, wave disturbances can spread across the grid and create larger wave patterns over time.

This demonstrates the main concept of Cellular Automata, where complex overall behaviour emerges from simple local rules applied repeatedly to neighbouring cells. In this model, wind forcing and small random disturbances are added so the water surface is continuously changing rather than remaining static. The CA is important because it provides a dynamic environmental condition for the USVs, allowing the simulation to study how changing wave behaviour may affect the performance of autonomous agents.

4. Results and Outcomes

The simulation was tested using wave strengths of 0.1, 1.0 and 2.0. All three runs successfully collected and delivered all generated plastic, with five transport trips completed in each run.

Wave strength	Finish tick	Trips	Generated kg	Delivered kg	SC idle %	SC patrol %	Mean |wave|	Missed sightings
0.1	956	5	48.3	48.3	20.8	39.0	0.08	8
1.0	1374	5	42.5	42.5	14.3	47.6	0.58	7
2.0	1378	5	41.0	41.0	23.6	36.4	1.18	70

As wave strength increased, the mean wave magnitude increased from 0.08 at wave strength 0.1 to 0.58 at 1.0 and 1.18 at 2.0. The completion time also increased from 956 ticks at wave strength 0.1 to 1374 ticks at 1.0 and 1378 ticks at 2.0. However, the total amount of generated plastic differed between runs, so completion time cannot be attributed only to wave strength.

The clearest effect of stronger waves was observed in the number of missed sightings. The low- and medium-wave runs recorded only 8 and 7 missed sightings, while the high-wave run recorded 70 missed sightings. This suggests that stronger CA-generated waves can significantly reduce the SC-USV's ability to detect plastic, particularly when plastic is temporarily hidden within wave troughs.

5. Discussion and Conclusion

The simulation shows that combining discrete agent-based modelling with cellular automata works well for representing both autonomous USV behaviour and a changing water environment. The SC-USV and T-USV cooperated to search for, collect, transfer and deliver all generated plastic in every run. The CA-generated waves affected both their movement and their sensing.

The results suggest that waves affect the system in two different ways. From low to medium wave strength, completion time per kilogram of plastic increased noticeably while missed sightings stayed almost the same, which points to the waves slowing the USVs down. At the highest wave strength, missed sightings rose sharply, but completion time barely changed. Since the waves keep moving, plastic hidden in a trough soon becomes visible again, so detection was delayed rather than lost. This suggests the cooperative system is fairly robust to temporary detection failures.

These findings should be treated as preliminary. Each wave strength was run only once, and the amount of plastic generated differed between runs. Repeating each setting with several random seeds and a fixed plastic layout would give more reliable results. Overall, the hybrid ABM–CA approach shows how autonomous agents and environmental conditions can interact within the same simulation. It provides a useful foundation for extending the model to more complex marine collection scenarios.